

Operator on Inner Product Spaces.

• $F = \mathbb{R}$ or $\mathbb{C}$.

• V & W denote finite-dim inner product spaces over F .

→ Adjoint, T^* : let $T \in L(V, W)$. The adjoint of $T: V \rightarrow W$ is the function $T^*: W \rightarrow V$ s.t.

$$\langle T\bar{v}, \bar{w} \rangle = \langle \bar{v}, T^*\bar{w} \rangle \quad \forall \bar{v} \in V, \forall \bar{w} \in W.$$

→ Consider a linear functional $\varphi(\bar{v}) = \langle T\bar{v}, \bar{w} \rangle, \bar{v} \in V, \varphi: V \rightarrow F$. $\exists$ a unique vector $T^*\bar{w} \in V$ s.t. $\varphi(\bar{v}) = \langle \bar{v}, T^*\bar{w} \rangle, \forall \bar{v} \in V$.

Examples: ① $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2, T(x_1, x_2, x_3) = (x_2 + 3x_3, 2x_1)$.

$$\langle (x_1, x_2, x_3), T^*(y_1, y_2) \rangle = \langle T(x_1, x_2, x_3), (y_1, y_2) \rangle$$

$$= \langle (x_2 + 3x_3, 2x_1), (y_1, y_2) \rangle$$

$$= x_2 y_1 + 3x_3 y_1 + 2x_1 y_2 = \langle (x_1, x_2, x_3), (2y_2, y_1, 3y_1) \rangle$$

$$\Rightarrow T^*(y_1, y_2) = (2y_2, y_1, 3y_1)$$

② $\bar{u} \in V, \bar{x} \in W, T \in L(V, W); T\bar{v} = \langle \bar{v}, \bar{u} \rangle \bar{x}, \forall \bar{v} \in V$.

For a fixed, arbitrary $\bar{w} \in W$,

$$\langle \bar{v}, T^*\bar{w} \rangle = \langle T\bar{v}, \bar{w} \rangle = \langle \langle \bar{v}, \bar{u} \rangle \bar{x}, \bar{w} \rangle$$

$$= \langle \bar{v}, \bar{u} \rangle \langle \bar{x}, \bar{w} \rangle = \langle \bar{v}, \langle \bar{x}, \bar{w} \rangle \bar{u} \rangle$$

$$\Rightarrow T^*\bar{w} = \langle \bar{x}, \bar{w} \rangle \bar{u} \quad \forall \bar{w} \in W.$$

Thm. If $T \in L(V, W)$, then $T^* \in L(W, V)$.

Proof: $\forall c \in F, \forall \bar{v} \in V, \forall \bar{w}_1, \bar{w}_2 \in W$,

$$\langle \bar{v}, T^*(c\bar{w}_1 + \bar{w}_2) \rangle = \langle \bar{v}, c(T^*\bar{w}_1) + T^*\bar{w}_2 \rangle$$

$$\Rightarrow T^*(c\bar{w}_1 + \bar{w}_2) = cT^*\bar{w}_1 + T^*\bar{w}_2$$

Thm Properties of the adjoint:

(a) $(S+T)^* = S^* + T^* \quad \forall S, T \in L(V, W);$

(b) $\lambda T^* = \bar{\lambda} T^* \quad \forall \lambda \in F, \forall T \in L(V, W);$

(c) $(T^*)^* = T \quad \forall T \in L(V, W);$

(d) $1^* = 1$, 1 is the identity op. on V .

(e) $(ST)^* = T^* S^* \quad \forall T \in L(V, W) \text{ and } S \in L(W, U)$
inner product space over F .

Proof: _____

Orthogonal complement: If $U \subset V$, then the orthogonal complement of U , denoted as $U^\perp$, is the set of all vectors $v \in V$ that are orthogonal to every vector in U :

$$U^\perp = \{v \in V : \langle v, u \rangle = 0 \quad \forall u \in U\}.$$

Thm: let $T \in L(V, W)$. Then

(a) $\text{null } T^* = (\text{range } T)^\perp$

(b) $\text{range } T^* = (\text{null } T)^\perp$

(c) $\text{null } T = (\text{range } T^*)^\perp$

(d) $\text{range } T = (\text{null } T^*)^\perp$

Proof: (a) let $\bar{w} \in W$, $\bar{w} \in \text{null } T^*$
 $\Leftrightarrow T^* \bar{w} = \bar{0} \Leftrightarrow \langle v, T^* \bar{w} \rangle = 0 \quad \forall v \in V$
 $\Leftrightarrow \langle T v, \bar{w} \rangle = 0 \quad \forall v \in V$
 $\Leftrightarrow \bar{w} \in (\text{range } T)^\perp$
 $\therefore \text{null } T^* = (\text{range } T)^\perp$

✓ The conjugate transpose of an $m \times n$ matrix is the $n \times m$ matrix obtained by interchanging the rows and columns and then taking the complex conjugate of each entry.

→ The adjoint of a linear map does not depend on a choice of basis.

→ W.r.t. nonorthonormal bases, the matrix of T^* does not necessarily equal to the conjugate transpose of the matrix of T .

Thm. Let $T \in L(V, W)$. Suppose $\bar{e}_1, \dots, \bar{e}_n$ is an orthonormal basis of V and $\bar{f}_1, \dots, \bar{f}_m$ is an orthonormal basis of W . Then,

$M(T^*, (\bar{f}_1, \dots, \bar{f}_m), (\bar{e}_1, \dots, \bar{e}_n))$
is the conjugate transpose of
 $M(T, (\bar{e}_1, \dots, \bar{e}_n), (\bar{f}_1, \dots, \bar{f}_m))$.

Proof: $B_V = \bar{e}_1, \dots, \bar{e}_n$, $B_W = \bar{f}_1, \dots, \bar{f}_m$

j^{th} column of $[T]_{B_V, B_W} = [T \bar{e}_j]_{B_W}$

$$T \bar{e}_j = \sum_{i=1}^m \langle T \bar{e}_j, \bar{f}_i \rangle \bar{f}_i. \quad [M(T)]_{i,j} = \langle T \bar{e}_j, \bar{f}_i \rangle.$$

$$\begin{aligned} [M(T^*)]_{i,j} &= \langle T^* \bar{f}_j, \bar{e}_i \rangle = \overline{\langle \bar{e}_i, T^* \bar{f}_j \rangle} \\ &= \overline{\langle T \bar{e}_i, \bar{f}_j \rangle} = [M(T)]_{j,i}. \end{aligned}$$

→ An operator $T \in L(V)$ is called self-adjoint
if $T = T^*$. That is, $T \in L(V)$ is self-adjoint
iff $\langle T\bar{v}, \bar{w} \rangle = \langle \bar{v}, T\bar{w} \rangle \quad \forall \bar{v}, \bar{w} \in V$.

Notice that for self-adjoint operators $T_1, T_2 \in L(V)$,
 $T_1 + T_2$ is also self-adjoint
and $\forall c \in \mathbb{R}, cT_1$ is also self-adjoint.

Thm Eigenvalues of self-adjoint operators
are real always.

Or Every eigenvalue of a self-adjoint ϕ .
is real.

Proof: ~~Let $\{\lambda, \bar{v}\}$ be an eigenvalue and v~~
for a self-adjoint operator $T \in L(V)$,
let λ be an eigenvalue & $\bar{v}$ be its
corresponding eigenvector.

$$\begin{aligned}\langle T\bar{v}, \bar{v} \rangle &= \lambda \langle \bar{v}, \bar{v} \rangle = \langle \bar{v}, T^*\bar{v} \rangle = \langle \bar{v}, T\bar{v} \rangle = \langle \bar{v}, \lambda \bar{v} \rangle \\ &= \bar{\lambda} \langle \bar{v}, \bar{v} \rangle\end{aligned}$$

$$\Rightarrow \lambda = \bar{\lambda} \Rightarrow \lambda \in \mathbb{R}.$$

Thm. let V be a complex inner product space & $T \in L(V)$. Suppose $\langle T\bar{v}, \bar{v} \rangle = 0 \quad \forall \bar{v} \in V$. Then $T=0$.

Proof: $\langle T\bar{u}, \bar{w} \rangle$

$$= \frac{1}{4} \left[\langle T(\bar{u} + \bar{w}), \bar{u} + \bar{w} \rangle - \langle T(\bar{u} - \bar{w}), \bar{u} - \bar{w} \rangle + i \left(\langle T(\bar{u} + i\bar{w}), \bar{u} + i\bar{w} \rangle - \langle T(\bar{u} - i\bar{w}), \bar{u} - i\bar{w} \rangle \right) \right]$$

$\forall \bar{u}, \bar{w} \in V$.

Thm. let V be a complex inner product space & $T \in L(V)$. Then T is self-adjoint iff $\langle T\bar{v}, \bar{v} \rangle \in \mathbb{R} \quad \forall \bar{v} \in V$.

Proof: let $\bar{v} \in V$,

$$\begin{aligned} \langle T\bar{v}, \bar{v} \rangle - \overline{\langle T\bar{v}, \bar{v} \rangle} &= \langle T\bar{v}, \bar{v} \rangle - \langle \bar{v}, T\bar{v} \rangle \\ &= \langle T\bar{v}, \bar{v} \rangle - \langle T^* \bar{v}, \bar{v} \rangle = \langle (T - T^*)\bar{v}, \bar{v} \rangle. \end{aligned}$$

Thm. let T be a self-adjoint op. on V s.t. $\langle T\bar{v}, \bar{v} \rangle = 0 \quad \forall \bar{v} \in V$. Then $T=0$.

Proof: Proved for $F = \mathbb{C}$ earlier.

$F = \mathbb{R}$, $\langle T\bar{u}, \bar{w} \rangle = \frac{1}{4} \left[\langle T(\bar{u} + \bar{w}), \bar{u} + \bar{w} \rangle - \langle T(\bar{u} - \bar{w}), \bar{u} - \bar{w} \rangle \right],$

as $\langle T\bar{w}, \bar{u} \rangle = \langle \bar{w}, T\bar{u} \rangle = \langle T\bar{u}, \bar{w} \rangle$

Normal operators

✓ An operator $T \in L(V)$ is called normal if $TT^* = T^*T$.

Alternatively, an operator on an inner product space is called normal if it commutes w/ its adjoint.

→ Whenever $T = T^*$, $TT^* = T^*T$; i.e., self-adjoint operators are normal.

Thm. $T \in L(V)$ is normal $\iff$
 $\|T\bar{v}\| = \|T^*\bar{v}\| \quad \forall \bar{v} \in V$.

Proof: $T^*T = TT^* \iff \langle (T^*T - TT^*)\bar{v}, \bar{v} \rangle = 0 \quad \forall \bar{v} \in V$
 $\iff \langle T^*T\bar{v}, \bar{v} \rangle = \langle TT^*\bar{v}, \bar{v} \rangle \quad \forall \bar{v} \in V$
 $\iff \langle T\bar{v}, T\bar{v} \rangle = \langle T^*\bar{v}, T^*\bar{v} \rangle \quad \forall \bar{v} \in V$.

$$\rightarrow (T^*T - TT^*)^* = T^*T - TT^*.$$

Thm. Let $T \in L(V)$ is normal & $\bar{v} \in V$ is an eigenvector of T w/ eigenvalue λ .

Then $\bar{v}$ is also an eigenvector of T^* corresponding to eigenvalue $\bar{\lambda}$.

Proof: $\because T$ is normal, $(T - \lambda \mathbb{1})$ is also normal.
 $\therefore \|(T - \lambda \mathbb{1})\bar{v}\| = \|(T - \lambda \mathbb{1})^*\bar{v}\| = \|(T^* - \bar{\lambda} \mathbb{1})\bar{v}\| = 0$.

$$\Rightarrow T^*\bar{v} = \bar{\lambda}\bar{v}.$$

Thm. Let $T \in L(V)$ is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof: Let $(\alpha, \bar{u})$ & $(\beta, \bar{v})$ be distinct pairs of eigenvalues & corresponding eigenvector.
 $T\bar{u} = \alpha\bar{u}$, $T\bar{v} = \beta\bar{v}$.

We get, $T^* \bar{v} = \bar{\beta} \bar{v}$.

$$\begin{aligned} \textcircled{\otimes} \textcircled{\otimes} (\alpha - \beta) \langle \bar{u}, \bar{v} \rangle &= \langle \alpha \bar{u}, \bar{v} \rangle - \langle \bar{u}, \bar{\beta} \bar{v} \rangle \\ &= \langle T\bar{u}, \bar{v} \rangle - \langle \bar{u}, T^* \bar{v} \rangle \\ &= 0 \end{aligned}$$

~~⊗~~ That is, either $\alpha = \beta$, but this is not possible,

$$\text{or, } \langle \bar{u}, \bar{v} \rangle = 0 \Rightarrow \bar{u} \perp \bar{v}.$$